that extracts parameter fragments of N (e.g., blocks of weights/biases or individual rows/columns). Fix an addressing function $\mathrm{addr} : \mathcal{J} \to [L_{\max}]$ that assigns each fragment index j to a prompt slot.

Lemma 7.3 (Encoding feasibility via coordinate splitting)**.** *Let D be the model width. Fix integers d_k, d_v with $D \geq d_k + d_v$. Choose $W_K, W_V \in \mathbb{R}^{D\times D}$ so that they read disjoint coordinate blocks, e.g.,*

$$W_K = \begin{bmatrix} I_{d_k} & 0 \\ 0 & 0 \end{bmatrix}, \qquad W_V = \begin{bmatrix} 0 & 0 \\ 0 & I_{d_v} \end{bmatrix}$$

(after padding to $D \times D$). Let each prompt token be a row *vector $p_i \in \mathbb{R}^{1\times D}$ and define*

$$p_i = \left[\, \kappa(i)^\top \;\; \nu(y_i)^\top \;\; 0^\top \,\right] \in \mathbb{R}^{1\times D},$$

where $\kappa(i) \in \mathbb{R}^{d_k}$ and $\nu(y_i) \in \mathbb{R}^{d_v}$ are prescribed. Then $(p_i W_K)_{1:d_k} = \kappa(i)$ and $(p_i W_V)_{(d_k+1):(d_k+d_v)} = \nu(y_i)$ simultaneously.

Definition 7.4 (Encoding map)**.** Fix an addressing (key) scheme $\kappa : [L_{\max}] \to \mathbb{R}^{d_k}$ and a value embedding map $\nu : \mathbb{R}^q \to \mathbb{R}^{d_v}$. Assuming Lemma 7.3, define the encoding $p_N = \mathrm{Enc}(N) \in \mathbb{R}^{L_{\max}\times D}$ as follows: for each $j \in \mathcal{J}$, let $i = \mathrm{addr}(j)$ and set the value stored at slot i to be $\nu(\mathrm{payload}_j(N))$, while the slot key is fixed as $\kappa(i)$.

Remark 7.5. The addressing (key) scheme is fixed across tasks, and the task-specific freedom enters only through the stored values (payloads). Thus, a central design variable becomes *when* and *which* slots are read during execution.

7.2. Execution: Layerwise Emulation

Definition 7.6 (Registers via coordinate projections)**.** Assume the model width D is large enough to decompose $\mathbb{R}^D$ into disjoint coordinate blocks

$$\mathbb{R}^D \cong \mathbb{R}^{r_0} \oplus \mathbb{R}^{r_1} \oplus \cdots \oplus \mathbb{R}^{r_M}.$$

For $\ell \in [M]$, define $\mathrm{Read}_\ell(z)$ as the projection onto block ℓ, and define $\mathrm{Write}_\ell(z, v)$ as the operation that replaces block ℓ of z by v (keeping all other blocks unchanged).

Definition 7.7 (Layer-emulation macro-step)**.** Let the ideal layer update at depth ℓ be

$$G_\ell(u) = \sigma\big(W^{(\ell)} u + b^{(\ell)}\big),$$

where u denotes the current hidden vector stored in designated registers of the work token. We denote by $\widetilde{G}_\ell$ the approximate map implemented by the Transformer by (i) retrieving the required parameter chunks from the prompt via routing and (ii) applying an FFN-based local update that depends on (retrieved fragments, current registers).

Proposition 7.8 (Interpreter stage)**.** *Fix an MLP depth S and parameter bound Λ. For any $\eta > 0$, there exist model sizes $(T, D, \text{FFN width})$ and a temperature $\tau > 0$, and there exist fixed parameters θ^* within the simplified Transformer family such that the following holds:*

for every ReLU MLP N satisfying Definition 7.1, there exists a prompt $p_N = \mathrm{Enc}(N) \in \mathbb{R}^{L_{\max}\times D}$ (using zero-padding if needed) such that

$$\sup_{x\in\mathcal{X}} \left| F_{\theta^*}(p_N, x) - N(x) \right| \leq \eta.$$

Proof sketch. We implement the forward pass of N via macro-steps for layers $\ell = 1, \ldots, S-1$. At macro-step ℓ, the work token forms queries so that, by the key design κ, the correct slot attains an $(i^\star, \Delta)$-margin. Lemma 6.2 then yields exponentially accurate retrieval, and it suffices to pick τ so that $(L_{\max} - 1)e^{-\Delta/\tau} \leq \eta_{\mathrm{route}}$.

On the resulting compact set of register states and retrieved fragments (from bounded parameters and compact $\mathcal{X}$), Lemma 6.6 gives a token-wise FFN approximating the intended local update map G_ℓ within the arithmetic budget. Finally, Lemma 6.8 allocates per-step budgets so that routing and arithmetic errors accumulate to at most η, yielding $\sup_{x\in\mathcal{X}} \left| F_{\theta^*}(p_N, x) - N(x) \right| \leq \eta$. □

Remark 7.9 (Putting errors together). For any $g \in C(\mathcal{X})$, let N_g satisfy $\|N_g - g\|_\infty < \varepsilon/2$ by the standard UAT for ReLU MLPs. Applying Proposition 7.8 with $\eta = \varepsilon/2$ gives

$$\begin{aligned} \|F_{\theta^*}(p_{N_g}, \cdot) - g\|_\infty &\leq \|F_{\theta^*}(p_{N_g}, \cdot) - N_g\|_\infty \\ &\quad + \|N_g - g\|_\infty \\ &< \varepsilon. \end{aligned}$$

which implies Theorem 5.1.

8. Discussion and Limitations

Discussion. We position prompt engineering as a *theoretical object*: for a *fixed* backbone, varying the prompt induces a well-defined hypothesis class. This reframes the prompt as an external parameter—a program input—so expressivity questions can be posed directly about a single mathematical object.

Our contribution is constructive and mechanism-oriented. The proof isolates a reusable decomposition of prompt-based switching: prompt slots serve as addressable memory, attention performs *routing* via near-selective retrieval, the FFN performs *arithmetic* via local computation conditioned on retrieved fragments, and depth implements *composition* by accumulating these updates into multi-stage computation. Consequently, the construction makes explicit *what* is stored, *how* it is selected, and *how* it is transformed across layers.

This viewpoint also clarifies where practical constraints enter—memory via L, routing reliability via Δ/τ, and arith-

metic accuracy via FFN width—even though we do not optimize these costs here. By turning informal prompt "tips" into explicit knobs, it sets up subsequent work to quantify scaling laws and lower bounds under realistic constraints.

Limitations. Our theorem is existential and concerns expressivity; it should not be read as an empirical claim that pretrained LLMs satisfy Prompt-UAT as stated. Moreover, while we prove that an appropriate prompt exists for each target, we do not provide efficient procedures for finding such prompts. We also do not quantify the costs induced by realistic constraints—such as prompt length, finite precision, norm bounds, or the margins/temperature scaling required for reliable routing—within the main text.

Future Work. A direct next step is to quantify trade-offs between prompt length/precision/norm constraints and approximation error ε. A second direction is to derive lower bounds and impossibility results for hard prompts (finite vocabularies and finite lengths), including information-theoretic limits on how much task specification can be conveyed. A third direction is to connect existence to *learnability*: whether the existential prompts can be recovered by practical optimization, and how pretrained constraints (architecture, training-induced biases, and optimization dynamics) alter the boundary between possible and impossible switching.

9. Conclusion

We introduced a framework that treats prompt engineering as a coherent theoretical object rather than an empirical craft. Instead of changing weights across tasks, we ask what functionality can be switched when a single backbone is kept fixed as an executor and the prompt is used as an external program. We formalize the family of behaviors obtained by varying prompts under a fixed backbone and prove a constructive existential guarantee: by changing only the prompt, the fixed executor can approximate a broad class of targets to arbitrary accuracy. The construction makes the switching mechanism explicit—prompt slots act as addressable memory, attention provides selective routing, the FFN provides local arithmetic, and depth-wise repetition composes these updates into a full computation. This viewpoint motivates future work on trade-offs under length, precision, and norm constraints, limits for finite hard prompts, pretraining-induced restrictions, and the learnability of effective prompts under practical optimization.

Impact Statement

We study prompt engineering as an expressivity problem: what a *fixed* backbone can realize when only the prompt is varied. Our impact is primarily conceptual and theoretical. By formalizing the prompt-induced hypothesis class and proving a constructive existence result for prompt-based switching, we provide precise language for reasoning about what prompts can enable or restrict in principle, and for separating expressivity questions from optimization or training effects.

This work is foundational rather than deployment-facing. We do not claim that pretrained large language models satisfy the stated property, and we do not provide procedures for finding effective prompts under realistic constraints such as length, precision, or norm bounds. Accordingly, the results should not be interpreted as evidence of practical capability, nor as any form of safety, robustness, or reliability guarantee for prompt-only control.

Potential downstream benefits include improved interpretability of prompt-conditioned behavior, clearer theoretical guidance for prompt tuning and prompt search, and more principled characterization of failure modes and invariants in prompt-controlled systems. Potential risks include misreading the existence result as a practical capability claim or using it to justify overconfident reliance on prompt-only control in safety-critical settings. We mitigate these risks by clearly delimiting the scope of the theorem, emphasizing the absence of constraint-aware and learnable guarantees, and explicitly identifying robustness, constrained trade-offs, and learnability as open directions.

References

Akyürek, E., Schuurmans, D., Andreas, J., Ma, T., and Zhou, D. What learning algorithm is in-context learning? investigations with linear models. *arXiv preprint arXiv:2211.15661*, 2022.

Bournez, O., Cohen, J., and Wurm, A. A universal uniform approximation theorem for neural networks. In *50th International Symposium on Mathematical Foundations of Computer Science (MFCS 2025)*, pp. 29–1. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2025.

Cheng, X., Chen, Y., and Sra, S. Transformers implement functional gradient descent to learn non-linear functions in context. *arXiv preprint arXiv:2312.06528*, 2023.

Conmy, A., Mavor-Parker, A., Lynch, A., Heimersheim, S., and Garriga-Alonso, A. Towards automated circuit discovery for mechanistic interpretability. *Advances in Neural Information Processing Systems*, 36:16318–16352, 2023.

Crosbie, J. and Shutova, E. Induction heads as an essential mechanism for pattern matching in in-context learning. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pp. 5034–5096, 2025.

Dai, D., Sun, Y., Dong, L., Hao, Y., Ma, S., Sui, Z., and Wei, F. Why can gpt learn in-context? language models secretly perform gradient descent as meta-optimizers. In *Findings of the Association for Computational Linguistics: ACL 2023*, pp. 4005–4019, 2023.

Ding, N., Qin, Y., Yang, G., Wei, F., Yang, Z., Su, Y., Hu, S., Chen, Y., Chan, C.-M., Chen, W., et al. Parameter-efficient fine-tuning of large-scale pre-trained language models. *Nature machine intelligence*, 5(3):220–235, 2023.

Furuya, T., Puthawala, M., Lassas, M., and De Hoop, M. V. Globally injective and bijective neural operators. *Advances in Neural Information Processing Systems*, 36: 57713–57753, 2023.

Furuya, T., de Hoop, M. V., and Peyré, G. Transformers are universal in-context learners. *arXiv preprint arXiv:2408.01367*, 2024a.

Furuya, T., Taniguchi, K., and Okuda, S. Quantitative approximation for neural operators in nonlinear parabolic equations. *arXiv preprint arXiv:2410.02151*, 2024b.

Garg, S., Tsipras, D., Liang, P. S., and Valiant, G. What can transformers learn in-context? a case study of simple function classes. *Advances in neural information processing systems*, 35:30583–30598, 2022.

Geva, M., Bastings, J., Filippova, K., and Globerson, A. Dissecting recall of factual associations in auto-regressive language models. *arXiv preprint arXiv:2304.14767*, 2023.

Hendel, R., Geva, M., and Globerson, A. In-context learning creates task vectors. *arXiv preprint arXiv:2310.15916*, 2023.

Jain, A., Chaudhuri, S., Reps, T., and Jermaine, C. Prompt tuning strikes back: Customizing foundation models with low-rank prompt adaptation. *Advances in Neural Information Processing Systems*, 37:47297–47316, 2024.

Jiang, H. and Li, Q. Approximation rate of the transformer architecture for sequence modeling. *Advances in Neural Information Processing Systems*, 37:68926–68955, 2024.

Kim, N., Min, C., and Park, S. Minimum width for universal approximation using relu networks on compact domain. *arXiv preprint arXiv:2309.10402*, 2023.

Kovachki, N., Li, Z., Liu, B., Azizzadenesheli, K., Bhattacharya, K., Stuart, A., and Anandkumar, A. Neural operator: Learning maps between function spaces with applications to pdes. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

Le, M., Nguyen, C., Nguyen, H., Tran, Q., Le, T., and Ho, N. Revisiting prefix-tuning: Statistical benefits of reparameterization among prompts. *arXiv preprint arXiv:2410.02200*, 2024.

Li, C., Liu, X., Zhang, Z., Wang, Y., Liu, C., Lan, Y., and Shen, C. Concentrate attention: Towards domain-generalizable prompt optimization for language models. *Advances in Neural Information Processing Systems*, 37: 3391–3420, 2024.

Lin, Y.-C., Li, W.-H., Chen, J.-C., and Chen, C.-S. Accept: Adaptive codebook for composite and efficient prompt tuning. *arXiv preprint arXiv:2410.12847*, 2024.

Liu, Z., Wang, Y., Vaidya, S., Ruehle, F., Halverson, J., Soljačić, M., Hou, T. Y., and Tegmark, M. Kan: Kolmogorov-arnold networks. *arXiv preprint arXiv:2404.19756*, 2024.

Merrill, W. and Sabharwal, A. The expressive power of transformers with chain of thought. *arXiv preprint arXiv:2310.07923*, 2023a.

Merrill, W. and Sabharwal, A. The parallelism tradeoff: Limitations of log-precision transformers. *Transactions of the Association for Computational Linguistics*, 11:531–545, 2023b.

Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhardt, J. Progress measures for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217*, 2023.

Petrov, A., Torr, P. H., and Bibi, A. Prompting a pretrained transformer can be a universal approximator. *arXiv preprint arXiv:2402.14753*, 2024.

Rubin, O. and Berant, J. Retrieval-pretrained transformer: Long-range language modeling with self-retrieval. *Transactions of the Association for Computational Linguistics*, 12:1197–1213, 2024.

Sanford, C., Hsu, D. J., and Telgarsky, M. Representational strengths and limitations of transformers. *Advances in Neural Information Processing Systems*, 36:36677–36707, 2023.

Siegel, J. W. Optimal approximation rates for deep relu neural networks on sobolev and besov spaces. *Journal of Machine Learning Research*, 24(357):1–52, 2023.

Takakura, S. and Suzuki, T. Approximation and estimation ability of transformers for sequence-to-sequence functions with infinite dimensional input. In *International Conference on Machine Learning*, pp. 33416–33447. PMLR, 2023.